A Full Custom Design Methodology for n×n 6T SRAM Array with 2×2 Array Implementation

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***Abstract*—This work proposes a methodology for designing an n×n 6-Transistor (6T) Static Random Access Memory (SRAM) array by implementing a standard 6T SRAM followed by a 2×2 6T SRAM array in 90nm CMOS technology. The work demonstrates read, write, and hold operations of SRAM using transient and DC analysis in Cadence Virtuoso. It also discusses the design of peripheral circuitry, including a precharge circuit, a write driver, and a sense amplifier. Pre- and post-layout simulation results are also discussed to estimate the actual delay in accessing particular bits. An 18% increase in delay has been observed in post-layout simulations for 6T SRAM, while energy consumption remains nearly equal in both cases.**

***Index Terms*—6T SRAM, Static Noise Margin (SNM), SRAM Array, Sense Amplifier**

I. Introduction

Memory is considered a central part of digital systems [1]. Every modern system, such as processors, microcontrollers, or Field-Programmable Gate Arrays (FPGAs), uses memory to store data and other information. It is observed that processor sizes have decreased over the past couple of years, but memory has not been downscaled at a similar pace [2]. As a result, memory is not optimised for area and power and is therefore not compatible with next-generation computing. Moreover, memory access speed is relatively slow compared to the Central Processing Unit (CPU). Therefore, a significant amount of time is required to access the memory. Area is also considered a major concern because memory occupies 70 - 80 % of the area in many applications [3].

Memory is mainly classified into volatile and non-volatile memories. Non-volatile memories are slower but power-efficient, like Read-Only Memory (ROM) and Flash memory. In contrast, volatile memories, such as SRAM and Dynamic Random Access Memory (DRAM), offer high speed [4].

SRAM stores digital data using a pair of cross-coupled inverters [5]. The main advantage of SRAM over DRAM is that it does not require frequent refresh [1]. It stores data as long as the power supply is on; hence, it is preferred when performance is the target, such as in a System-on-Chip (SoC) or as CPU cache memory [6].

Several variants of SRAM are available, such as 6T, 8-transistor (8T), 10-transistor (10T), and so on, based on key performance metrics like area, power consumption, access time, and stability [5]. The 8T and 10T SRAMs have better stability but consume more area and power than the 6T SRAM [7]. The 6T is considered when area optimisation is the primary concern for applications requiring miniaturisation. It occupies the least area due to its fewer transistors [6]. This work presents the complete design methodology for building an SRAM array for educational purposes. The key contributions of the work are as follows:

- A full-custom design methodology for implementing an n × n 6T SRAM array with peripheral circuits is provided.
- A 6T SRAM and 2×2 SRAM array are designed, and the pre- and post-layout simulation results are discussed.
- Performance comparison with the existing work is carried out for 6T SRAM as well as a 2×2 array.

The rest of the paper is organised as follows: Section II discusses the standard 6T SRAM cell, including various operations, transistor sizing, cell ratio, and SNM. The methodology for designing an n×n array, followed by the case study of a 2×2 array, is described in Section III. Section IV discusses the simulation results of the 6T cell and the 2×2 array. The comparison with the existing work is also provided. Section V is about the conclusion of the work carried out and the future expansion of the work.

II. 6T SRAM Cell

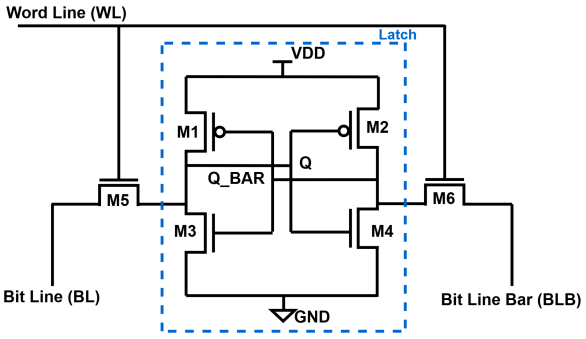


Fig. 1: 6T SRAM Cell

The standard 6T SRAM cell shown in (Fig. 1) consists of six MOSFETs: M1 and M2 (PMOS pull-up transistors), M3 and M4 (pull-down transistors), and M5 and M6 (NMOS access transistors). M1 and M3 form the first CMOS inverter, and

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M2 and M4 form the second inverter of the latch. The outputs are cross-coupled at nodes Q and Q_BAR, thereby creating a stable state that holds the data. The word line (WL) enables access to the pass transistors (M5 and M6) for read and write operations. The Bit Line (BL) and Bit Line Bar (BLB) are used to transfer data to and from the SRAM cell.

A. Operating Modes

SRAM operates in 3 modes *i.e.* hold, read, and write as explained below.

1) Hold Mode: This mode holds the values of the stored data. When WL is '0', transistors M5 and M6 (refer to Fig. 1) are turned off, which isolates the cross-coupled inverter from external interference and prevents any changes to the stored binary data.

2) Read Mode: For the read operation, BL and BLB are to be precharged to VDD. Now, make WL = '1'. The side that stores '0' (Q or Q_BAR) will discharge its bit line (BL or BLB) through the access transistor (M5 or M6) and pull-down transistors (M3 or M4). Both bit lines are connected to the sense amplifier (acts as a comparator, as shown in Fig. 5) that can sense the small change *i.e.* ΔV in bit lines. If BL $>$ BLB, the sense amplifier indicates '1' as output, or else it will give '0' as output, indicating '0' is stored inside the cell.

3) Write Mode: In write mode, the write driver (see Fig. 4) is used to force the BL and BLB to the desired values. Thereafter, make WL active high. The values will be stored in the inverters accordingly. For example, to write logic high into a cell, set BL = '1' (BLB = '0'). After this, set WL = '1'; this will turn on M5 and M6 (see Fig. 1), resulting in Q = '1'. Q is the input of inverters M2 and M4. It will give output Q_BAR as '0, meaning '1' (Q = '1') will be stored in the latch. Now turn off the WL. Similarly, we can explain the write operation for logic '0'. To achieve proper operation and SNMs, the SRAM transistors should be properly sized.

B. Transistor Sizing and Cell Ratio

In SRAM, data stability is affected by transistor sizing. The *cell ratio* (CR) controls read stability and is given by [8]:

$$CR = \frac{(W/L)_{PD}}{(W/L)_{PG}} \tag{1}$$

where, $(W/L)_{PD}$ and $(W/L)_{PG}$ represent the aspect ratios of pull-down transistors (M3, M4) and access transistors (M5, M6), respectively. A high CR value limits the voltage rise at the storage node during the read operation. Typical designs have CR $\geq$ 2.

The *pull-up ratio* (PR) regulates write-ability and is given as below [8]:

$$PR = \frac{(W/L)_{PU}}{(W/L)_{PG}} \tag{2}$$

where, $(W/L)_{PU}$ is the aspect ratio of pull-up transistors (M1, M2). A lower PR (weaker pull-up) makes it easier to flip the cell during a write operation. Typically, PR $\leq$ 1 is used, but for better SNM, we should set PR = 1 in read mode.

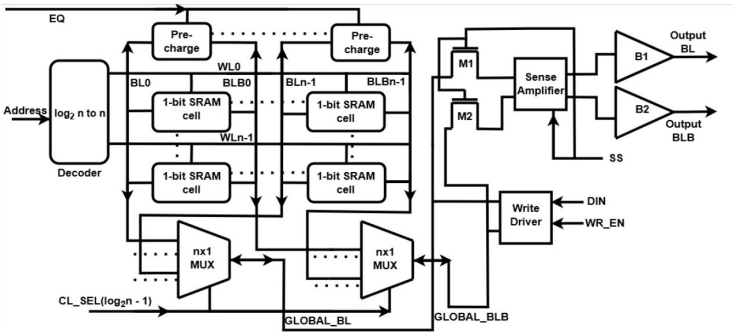


Fig. 2: Block Diagram of n×n 6T SRAM Array

C. Static Noise Margin

The Static Noise Margin refers to the maximum amount of DC noise voltage that an SRAM cell can withstand without changing its stored bit or value. It can be found by making the butterfly curve and identifying the side length of the largest square that can fit inside the VTC lobes:

$$\text{SNM} = \min(\text{side of square 1, side of square 2}) \tag{3}$$

According to the operating modes of SRAM, the SNM can be defined as follows:

1) Hold SNM: It is estimated while keeping WL at an active low. Access transistors (M5 and M6 (*refer to Fig. 1*)) are disabled, and hence the cell latches the ideal DC points.
2) Read SNM: It is measured when WL is '1'. The access transistors (M5 & M6) are turned on, creating a resistive divider that raises the zero-side voltage (Q or Q_BAR).
3) Write SNM: The WL will be high. BL & BLB will overwrite the cell's previous data with new data.

III. $n \times n$ 6T SRAM Array

An n×n array of 6T SRAM consists of 'n' rows and 'n' columns *i.e.* a total n^2 individual 1-bit SRAM cells as shown in Fig. 2. Row and column decoding logic must be integrated to access any individual cell in the array. For building an array, various peripheral circuits [4], such as a write driver, a row decoder, column muxes, pass transistors, and a sense amplifier, are required. For an n×n array, one Sense Amplifier, one Write Driver, 'n' precharge circuits, two column multiplexers (muxes) of n×1, a decoder of $log_2 n$ to n, and additionally 2 pass-transistors (M1 and M2) (refer to Fig. 2) and two buffers (B1 and B2) for BL and BLB are required for stable operation. Based on the address bits, the decoder selects the corresponding row. Word lines (WL) control the rows, whereas the columns are controlled through column select (CL_SEL). Each column will have a precharge circuit, and BL (BL0 to BLn-1) and BLB (BLB0 to BLBn-1) are connected to it. The particular row will be selected using a decoder, and a particular column will be selected using an MUX through the CL_SEL column selection lines. These muxes are bidirectional and are made using pass transistors or transmission gates. The two n×1 multiplexers will produce two signals, GLOBAL_BL and GLOBAL_BLB. Both signals pass through pass transistors (M1 and M2) and are connected to the Sense Amplifier and

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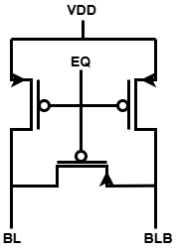


Fig. 3: Schematic of Precharge Circuit

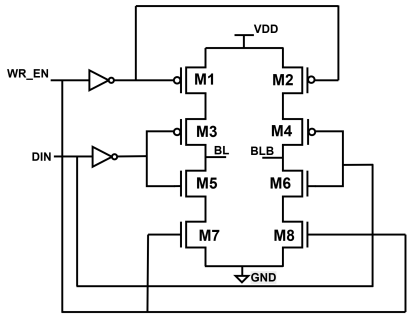


Fig. 4: Schematic of Write Driver

Write Driver. The gates of the pass transistors are connected to the SS (*i.e.* input signal to the sense amplifier). To reduce spikes during transient analysis, the sense amplifier output is passed through buffers (B1 and B2). DIN and WR_EN are the inputs to the write driver, where DIN is the input data, and WR_EN is the enable signal of the write driver, as shown in Fig. 3. EQ stands for equaliser, which is the signal for precharge. The functionality of the array can be understood through the peripheral circuits described below.

1) Precharge Circuit: The precharge circuit, as shown in Fig. 3, consists of three PMOS transistors operated by the EQ signal. BL and BLB are the outputs of the precharge circuit and are precharged to V_{DD} before each read operation to ensure proper sensing. When EQ is '0', BL and BLB will be precharged; when EQ is '1', no precharge will occur.

2) Write Driver: The write driver, as shown in Fig. 4, forces complementary data onto the bit lines, enabling a reliable write operation by overriding the stored value. When WR_EN is '1', only data can be written to the cell. Suppose DIN is kept '0'; then the '0' input will be given to the inverter formed by M4 & M6, and the input '1' will be given to the inverter formed by M3 & M5 (refer to Fig. 4), which makes BL = '0' & BLB = '1', ensuring proper signal generation in writing '0'.

3) Sense Amplifier: The sense amplifier [9] depicted in Fig. 5 detects the small voltage difference ΔV between bit lines (BL and BLB) during the read operation and amplifies this difference to V_{DD}. When SS (input signal of the sense amplifier) is active low, there is no direct path from VDD to ground (due to M7). BL (BLB) is available at node x (node y) via transistors M9 (M8). Before reading the value, BL and BLB are to be precharged to V_{DD}. Now, make SS = '1,' which activates the ground path through M7. Suppose the value of BL is 1.2 V and BLB = 1.1 V. The small difference between BL and BLB is induced by the value stored in the cell. The series

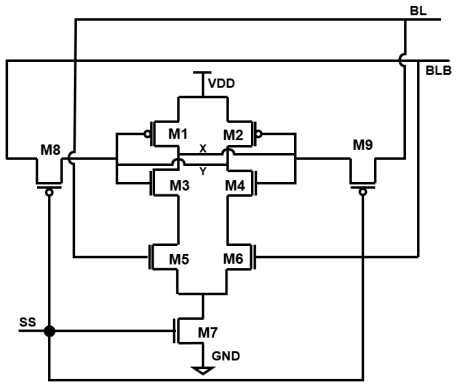


Fig. 5: Schematic of Sense Amplifier

transistor pair (M3 and M5) provides a strong pull-down path, causing node x to move towards ground. On the other hand, the voltage at node y becomes $\approx$ VDD due to the cross-coupled inverters formed by the M1-M3 and M2-M4 transistors in Fig. 5. In this manner, the nodes reach a stable value. So far, the peripheral circuits have been explained. Now, let us understand how an array works for read and write operations.

The read operation is carried out using a Sense Amplifier. The two muxes will take BL and BLB as inputs, and based on these inputs, GLOBAL_BL and GLOBAL_BLB will be determined. The sense amplifier will sense the difference, and the stored data (Q) will be read. The correct SRAM output during a read operation is ensured only when SS = '1'.

Write Driver is used to write data to SRAM cells. To write any data, the WR_EN signal in Fig. 2 must be active high, and the data will be given through DIN. To facilitate both read and write operations using common muxes, bidirectional muxes will be used. These muxes will operate in reverse direction in write mode, *i.e.*, the values of GLOBAL_BL and GLOBAL_BLB will change in response to the write driver's inputs. The CL_SEL, the select line for the muxes, selects the particular column in which the data is to be written. Accordingly, the BL and BLB values for the particular column will be changed using the muxes.

Table I shows the scalability of the design. It shows the number of cells required to implement an n$\times$n, 2$\times$2, and 4$\times$4 array. As the array size increases, the peripherals do not scale in proportion. By looking at this table, one can easily implement an m × n, n × m, or n × n SRAM array, where m is the number of rows (the total number of memory locations), and n is the number of columns. The implementation of an array is explained in the following section, using a 2$\times$2 array as a case study.

A. Working of 2×2 SRAM Array

Fig. 6 shows a 2$\times$2 6T SRAM comprising a 1x2 Decoder, 2$\times$2 (= 4) 6T SRAM cells, two 2x1 muxes, a Sense Amplifier, a Write Driver, and two precharge circuits, M1 and M2 pass transistors, and B1 and B2 buffers with different control signals (explained in Section III). The working of an array can be explained as follows: If we want to access the 1st row and 2nd column SRAM cell (refer to Fig. 6), select row 1

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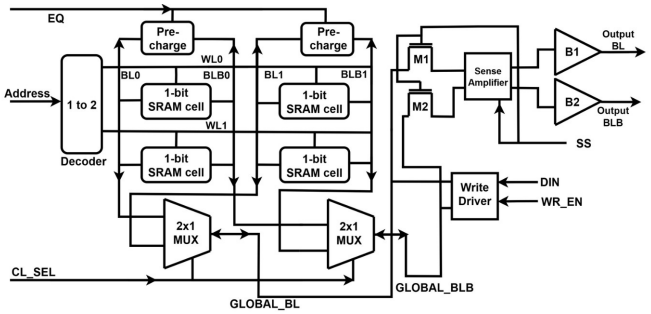


Fig. 6: Block Diagram of 2×2 SRAM Array

TABLE I: Scalability of Array

Component	**2×2**	**4×4**	$n \times n$
Decoder	$1 \to 2$	$2 \to 4$	$\log_2 n \to n$
Mux	$2(2 \times 1)$	$2(4 \times 1)$	$2(n \times 1)$
Mem. Cell	4	16	n^2
Pre-Charge Ckt.	2	4	n
Write Driver	1	1	1
Sense Amp.	1	1	1
Pass Transistors	2	2	2
Buffers	2	2	2

using a 1×2-decoder, which makes WL0 = '1'. To access the 2nd column, the bidirectional mux will initialise the BL1 and BLB1 signals. Depending on the read or write operation, the EQ signal will be set. (For example, make EQ = '0' for read operation). CL_SEL is set to '1' to select the second column mux. Thereafter, read or write operations can be performed as explained in Section III.

IV. Results and Discussions

To verify the functionality of the SRAM and array, transient and DC analyses are performed in Cadence Virtuoso using the Analog Digital Environment (ADE) L in 90nm CMOS technology. For the simulations, VDD is set to 1.2 V, and simulations are carried out at room temperature. Bitlines (BL and BLB) are precharged at 1.2 V before each read cycle.

A. 6T SRAM

1) Transient Analysis: Fig. 7 shows a transient response of a 6T SRAM, where the rectangle shows a write operation at 0.5×10^{-7} s. We can see that at this instant, BL = '1', BLB = '0', and WL = '1', which makes Q logic high and Q_BAR logic low. Similarly, the read operation can also be shown.

2) SNM Plot: The butterfly curve is used to capture the value of SNM. The SNM (given by Eq. (3)) of the hold operation is 299 mV, while for read and write operations, it is achieved as 125.3 mV and 284.5 mV, respectively. Eqs. (1) & (2) and the proper setting of the trip point (switching threshold of CMOS inverter) ensure the correct SNM values. Fig. 8 shows the read SNM curve for 6T SRAM, where the SNM value is lower than the hold SNM but still above the commonly accepted minimum value of 100 mV [10].

3) Pre- vs. Post-Layout Analysis: The propagation delay obtained from pre-layout or schematic simulation is calculated using the model parameters specified in the technology file.

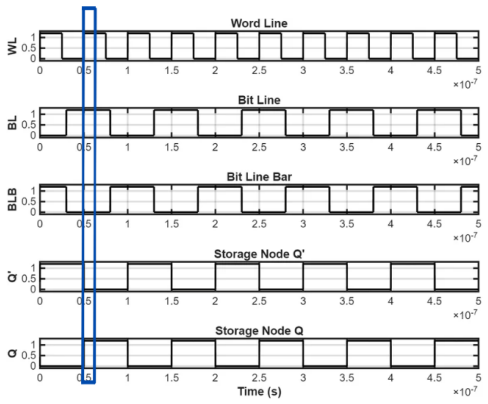


Fig. 7: Transient Response of 6T SRAM Cell

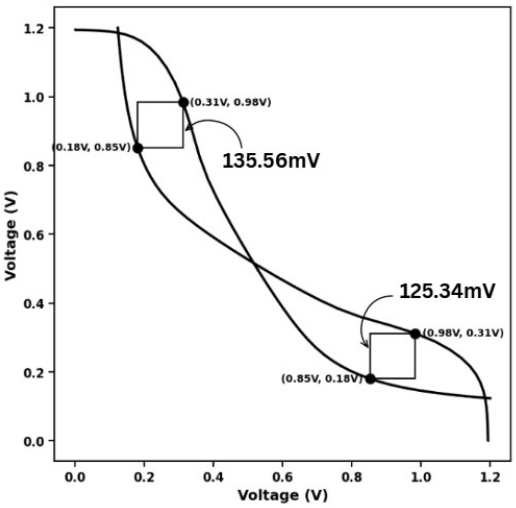


Fig. 8: Read SNM Plot for 6T SRAM cell

However, it does not account for the delay introduced by parasitics after layer placements. Therefore, the schematic is converted into physical layers, and parasitic extraction is performed to account for the delay introduced by the physical layers and interconnects in the layout. The layout of a 6T SRAM cell shown in Fig. 9 is designed in Cadence Virtuoso Layout XL and validated using the Assura tool. RC extraction is carried out using Assura QRC.

Table II compares key metrics, propagation delay, power, energy, and power-delay product, schematic and post-layout simulations of 6T SRAM. It can be seen that the delay has increased by 18% after post-layout simulation due to

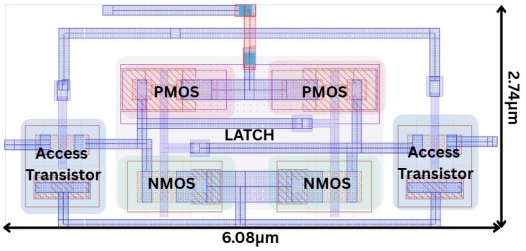


Fig. 9: Layout of 6T SRAM Cell

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TABLE II: Pre *vs.* Post-Layout Comparison of 6T SRAM

Parameter	Pre-layout	Post-layout	∼ Degradation (%)
Propagation Delay	54.25 ps	64.00 ps	18.10
Power	92.04 μW	91.9 μW	0.15
Energy	92.04 pJ	91.9 pJ	0.15
Power Delay Product	4.993 fs	5.882 fs	17.8

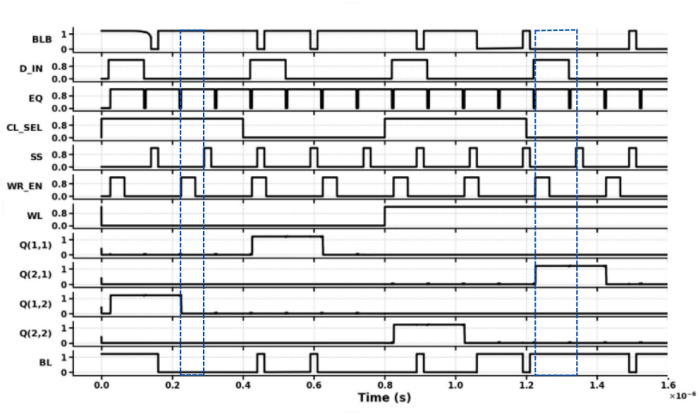


Fig. 10: Transient Response of 2×2 SRAM Array

interconnect parasitics.

B. 2×2 Array

1) Transient Analysis: Fig. 10 shows the transient response of a 2×2 6T SRAM array with control signals including EQ, SS, DIN, CL_SEL, WR_EN, WL (Address line), BL, and BLB. From the waveform above, it is observed that WL is kept low to access the first row and afterwards, it is kept high to access the second row. WR_EN is kept high for write operations, while DIN provides the input. The SS signal is active high during the read operation. It can also be observed that when CL_SEL is high, it accesses the second column, and when it is low, it accesses the first column.

Consider the dotted lines between 0.2 μs and 0.3 μs; we can see that DIN = '0'. CL_SEL = '1', which means the 2nd column will be selected, and WL = '0' means the 1st row is selected, which indicates cell (1,2) is selected. Since DIN = '0' means Q should be '0', we can see that Q changes from '1' to '0', after that instance, and when SS becomes '1', we can observe the BL = '0' and BLB = '1', reading the correct stored value for cell (1,2).

For another instance (see the dotted lines between 1.2 & 1.4 μs), we can see that CL_SEL = '0', which activates the 1st column, and WL is high, meaning the 2nd row is selected, so cell (2,1) will receive the value DIN = '1'(as WR_EN is high). From the Fig. 10, it can be seen that Q (2,1) is high at this point, indicating that the write operation is correct. Immediately after this, when SS is high, BL = '1' and BLB = '0', showing the correct read operation in the (2,1) cell.

2) Pre- vs. Post-Layout Analysis: The layout of a 2×2 6T SRAM array with peripheral circuits is shown in Fig. 11. It can be observed that the peripheral circuits occupy a major amount of area. Table III shows the pre- and post-layout comparison for the 2×2 array. Here, we observe a significant increase in access times after the post-layout simulation, primarily due to parasitics introduced by peripherals. However, this is not a major concern for larger arrays, as Table I shows that the number of peripheral circuits will not increase as much as the cell area does, and hence the delay they induce.

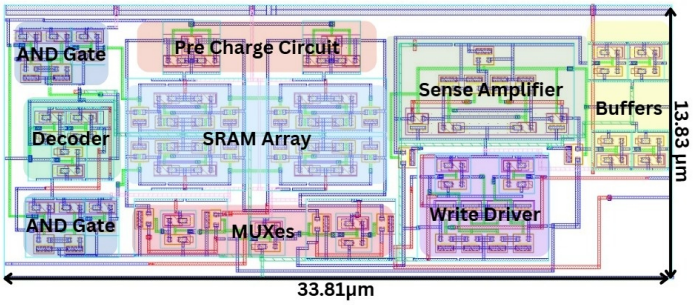


Fig. 11: Layout of 2×2 6T SRAM Array with Peripherals

TABLE III: Pre *vs.* Post-Layout Results of 2×2 Array

Metric	Pre-layout	Post-layout	∼ Degradation (%)
Write Access Time	0.56 ns	0.95 ns	67
Read Access Time	32.425 ps	54.30 ps	67.46
Power	11.28 μW	11.27 μW	∼0.088
Energy	18.048 pJ	18.032 pJ	0.088
Power Delay Product	6.30 ps	10.66 ps	69.15

TABLE IV: Comparison of 6T SRAM in 90 nm CMOS

Parameter	This Work	[1]	[5]	[6]	[11]
Supply VDD (V)	1.2	1.8	1.0	1.0	1.0
Read SNM (mV)	125.3	NR	NR	NR	85
Write SNM (mV)	284.5	NR	NR	NR	85
Hold SNM (mV)	299	NR	NR	NR	85
Read Power (μW)	4.58	1.49–2.39	10.59	558000	NR
Write Power (μW)	2.17	1.49–2.39	15.53	558000	NR
Hold Power (μW)	3.14	1.49–2.39	NR	558000	NR
Read Delay (ps)	53.65	26.99	NR	221000	96
Write Delay (ps)	53.65	4.23	NR	221000	300
Temperature (°C)	27	-50 to 250	NR	50–150	NR

NR = Not Reported.

C. Comparison with Existing Research

In this work, we presented a methodology for designing an n×n array for in-memory compute applications and demonstrated it using a 2×2 array. To validate the obtained results, we have compared our 6T SRAM and array with existing work in the literature while maintaining the same technology node, SRAM topology, and CMOS process. Recent papers that met all the criteria were difficult to find, so we drew on references from the older literature that operated at different supply voltages and had different array sizes.

Table IV compares the designed SRAM cell across supply voltage, different SNMs, read, write, and hold power consumption, read and write delays (access times), and operating temperature. From the table, we conclude that, among the other full-custom SRAM designs, our cell has the fastest read access time and operates at the highest frequency. Its SNM values are also better than those in [11] in all three cases: read, write, and hold.

Table V compares the performance of the array with the literature, including cell topology, memory size, supply voltage,

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TABLE V: Performance Comparison of SRAM Array in 90 nm CMOS Technology

Parameter	This Work	[12]	[13]	[14]	[15]
Cell Topology	6T	8T	6T	10T	6T
Memory Size	2×2	1 kB	32 kB	16 kB	64 kB
Supply VDD (V)	1.2	1.2	0.5	0.45	1.0
Read Power (μW)	∼11.28	21	3.27	NR	NR
Write Power (μW)	∼11.28	21	18.62	NR	NR
Read Access Time (ns)∗	0.03242	39.4	0.32	61.3	0.05
Write Access Time (ns)∗∗	0.5585	19.3	0.22	NR	0.14
Max. Frequency (GHz)	1.79	0.026	NR	0.3 @0.45 V	1.54
Cell Area (μm^2)	458.8	NR	(2.06 × 0.9)∗ ∗ ∗	NR	370000
Temperature ($^\circ$C)	27	27	−40 to 120	125	NR

NR = Not Reported. ∗ ∗ ∗ Single cell delay.

∗ Read access time is measured in the transition of SS (input of sense amplifier) from low to high with respect to the change in the outputs (BL and BLB).

∗∗ Write access time is measured in the transition of WR_EN (write enable) from low to high with respect to the change in Q and Q_BAR.

maximum operating frequency, read and write power, read and write access times, cell area, and temperature. From the table, we can say that our cell has the fastest read access time, 0.032 ns, compared to 0.05 ns in [16]. It also runs at the highest frequency, 1.79 GHz, compared to 1.54 GHz in [16]. In brief, our design performs well compared to existing work. Even though our SRAM cell is much smaller than the others, it still turns out to be faster and more stable in several cases.

V. Conclusion

A complete full-custom design methodology for an n×n SRAM array, including an example 2×2 SRAM array that can be scaled to any size, is presented in this work. To demonstrate the array's operation, pre- and post-layout simulation results are compared for both the single-cell 6T SRAM and the 2×2 array. A few percentage increase in access times (delays) is observed in both cases. The SNM values (read, hold, and write) are compared with the current literature and fall within an acceptable range.

Moving data between memory and the processor is a major bottleneck as AI applications continue to grow, and is known as the memory wall. It can be solved by performing the computation within the memory array, known as in-memory computing (CiM). Our 6T SRAM design is a natural fit for this, as it offers fast read access time and high operating frequency, making it well-suited to the frequent read operations that are a basic need of those CiM applications. Also, our SRAM Cell has good noise margins, ensuring it can reliably hold data even during computation. Although in this paper we have demonstrated a small 2×2 array, it can be scaled up to larger arrays for CiM applications [8], [17].

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